Practical 6: From KoboToolbox to CRD (No Coding Required)

Day 3 — Duration: 1.5 Hours

SDI Training for ICT Professionals — Climate Risk Database

What you will accomplish: By the end of this practical, your flood survey data from KoboToolbox will be visible on your CRD platform as an interactive map layer. You will do this using only QGIS and the CRD web interface — no programming or scripting required.

The workflow in 4 simple steps:

1. Export CSV from KoboToolbox
2. Import CSV into QGIS (turn coordinates into map points)
3. Export from QGIS as GeoJSON
4. Upload GeoJSON to CRD

Before you start: You need the test data from Practical 5 (3 flood survey submissions in KoboToolbox). Both KoboToolbox and your CRD should be running. QGIS should be installed.

Step 1: Export Data from KoboToolbox

Step 1.1: Open your browser and go to KoboToolbox:

```
http://localhost:8080
```

What this does: Opens the KoboToolbox web interface.

You should see: The KoboToolbox dashboard. Log in if needed.

Step 1.2: Click on your form: **"Flood Damage Assessment - Tanga 2026"**.

Step 1.3: Click on the **"Data"** tab at the top.

Step 1.4: Look for a **"Downloads"** or **"Export"** button. Click it.

You should see: An export options page where you can choose the format and settings for your data download.

Step 1.5: Configure the export:

- **Export type / Format:** Select **"CSV"** (Comma Separated Values)
- **Value and header format:** Select **"Labels"** (this gives you readable column names instead of codes)
- **Group separator:** Leave as default (/)
- Leave all other options as default

Step 1.6: Click **"Export"** or **"Create Export"**.

You should see: The export starts processing. After a few seconds, a download link appears.

Step 1.7: Click the **"Download"** link to save the CSV file.

You should see: A file downloads to your computer, named something like `Flood_Damage_Assessment_`

Step 1.8: Move or note where this file was saved (usually the `Downloads` folder). Remember this location — you will need it in the next step.

> **What is a CSV file?** CSV stands for Comma Separated Values. It is a simple text file where each line is a row of data, and commas separate the columns. You can open it in Excel or any spreadsheet program. For our purposes, the important columns are the GPS latitude and longitude.

Step 2: Open QGIS

Step 2.1: Open **QGIS Desktop** on your computer.

You should see: QGIS opens with a blank map canvas, a Layers panel on the left, and the Browser panel.

Step 2.2: It is good practice to save your work. Click **Project** → **Save As...** and save it as:

`kobo_to_crd_practical.qgz`

Save it on your Desktop or in your `sdi-training` folder.

Step 3: Import the CSV File into QGIS

This is the key step where we turn a spreadsheet of coordinates into map points.

Step 3.1: In the QGIS menu, click **Layer** → **Add Layer** → **Add Delimited Text Layer...**

You should see: The "Data Source Manager — Delimited Text" dialog opens.

Step 3.2: Click the **"..."** (browse) button next to **"File name"** and navigate to the CSV file you downloaded from KoboToolbox. Select it and click **"Open"**.

You should see: QGIS reads the file and shows a preview of the data at the bottom of the dialog. You should see your 3 submissions with all the columns.

Step 3.3: Configure the **File Format**:

- Make sure **"CSV (comma separated values)"** is selected
- Check **"First record has field names"** (this should be checked by default)

Step 3.4: Configure the **Geometry Definition** (this is the most important part):

- Select **"Point coordinates"**
- **X field (longitude):** Select the column containing longitude values. This is typically named:

```
_GPS_Location_of_Assessment_longitude
```

or similar (look for a column name containing "longitude" or "lon" or "X").

- **Y field (latitude):** Select the column containing latitude values. This is typically named:

```
_GPS_Location_of_Assessment_latitude
```

or similar (look for a column name containing "latitude" or "lat" or "Y").

Getting the right columns: KoboToolbox GPS questions create several columns: latitude, longitude, altitude, and precision. Make sure you select the **longitude** column for X and the **latitude** column for Y. A common mistake is swapping them — if your points appear in the wrong location, swap X and Y.

Step 3.5: Set the **Geometry CRS** (Coordinate Reference System):

- Click the small globe icon next to the CRS field
- In the search box, type: `4326`
- Select **"EPSG:4326 – WGS 84"** from the results
- Click **"OK"**

```
EPSG:4326 - WGS 84
```

What this does: Tells QGIS that the GPS coordinates are in the WGS 84 system, which is the standard GPS coordinate system used worldwide. All GPS devices and KoboToolbox use this system.

Step 3.6: Check the preview at the bottom of the dialog. You should see your data rows and the coordinate columns highlighted.

Step 3.7: Click **"Add"**.

You should see: The dialog stays open (in case you want to add more files). Your points should now appear on the QGIS map canvas.

Step 3.8: Click **"Close"** to close the dialog.

Step 4: Verify the Points on the Map

Step 4.1: Look at the QGIS map canvas. You should see 3 points plotted.

Step 4.2: If the points are too small to see, right-click the layer in the Layers panel → **"Zoom to Layer"**.

You should see: QGIS zooms in to show your 3 points. They should be located near Tanga, Tanzania.

Step 4.3: To confirm the location is correct, add a background map:

- In the Browser panel (left side), expand **"XYZ Tiles"**
- Double-click **"OpenStreetMap"**

You should see: An OpenStreetMap background appears. Your 3 points should be on or near Tanga city.

Step 4.4: Make sure the point layer is above the OpenStreetMap layer in the Layers panel. If not, drag it above.

Step 4.5: Click the **"Identify Features"** tool (the blue circle with an "i" in the toolbar) and click on one of the points.

You should see: A panel showing all the attributes for that point — flood depth, ward name, number of households, damage types, etc.

If your points appear in the ocean or wrong location:

- You may have swapped X (longitude) and Y (latitude). Remove the layer, go back to Step 3, and swap the X and Y column selections.
- Check that you selected EPSG:4326 as the CRS.

Step 5: Export as GeoJSON

CRD accepts GeoJSON files for upload. Let us convert the CSV-based layer into a proper GeoJSON file.

Step 5.1: In the QGIS Layers panel, right-click on your flood data layer (the point layer from the CSV).

Step 5.2: Click **"Export"** → **"Save Features As..."**.

You should see: The "Save Vector Layer as..." dialog.

Step 5.3: Configure the export settings:

- **Format:** Select **"GeoJSON"** from the dropdown
- **File name:** Click the **"..."** (browse) button, navigate to your Desktop, and name the file:

```
flood_assessment_tanga.geojson
```

- **CRS:** Make sure it says **"EPSG:4326 – WGS 84"**
- Leave all other settings as default

Step 5.4: Click **"OK"**.

You should see: A brief progress bar, then a success message. The file `flood_assessment_tanga.geojs` is now saved on your Desktop.

Step 5.5: Optionally, QGIS will ask "Add saved file to map?" Click **"Yes"** to see the exported layer.

Why GeoJSON? GeoJSON is an open standard format for geographic data. Unlike Shapefiles (which need 4+ files), GeoJSON is a single file. It is easy to share, upload, and works with virtually all GIS and web mapping tools.

Step 6: Upload GeoJSON to CRD

Step 6.1: Open your browser and go to your CRD:

```
http://localhost
```

Step 6.2: Make sure you are logged in as admin.

Step 6.3: Click **"Data"** → **"Upload Layer"** (or **"Upload Dataset"**).

Step 6.4: Drag the file `flood_assessment_tanga.geojson` from your Desktop into the upload area, or click **"Choose Files"** and select it.

Step 6.5: Click **"Upload"**.

You should see: A progress bar followed by a success message. The layer is now on your CRD.

Step 7: Set Metadata and Permissions

Step 7.1: After the upload completes, click **"Edit Metadata"** (or navigate to the layer's detail page and click the edit button).

Step 7.2: Fill in the metadata:

- **Title:** `Flood Damage Assessment - Tanga 2026`
- **Abstract:** `Field survey data from flood damage assessments conducted in Tanga City, Tanzania in March 2026. Data collected using KoboToolbox. Includes GPS locations, flood depth measurements, affected households, and damage type classifications.`
- **Keywords:** `flood`, `damage`, `assessment`, `tanga`, `kobotoolbox`, `field survey`
- **Category:** `Climatology / Meteorology` or `Disaster Risk`
- **Data source:** `KoboToolbox field survey`

Step 7.3: Click **"Save"**.

Step 7.4: Set permissions:

- Click **"Share"** or **"Permissions"**
- Set to **public view** and **public download**
- Click **"Save"**

Step 8: View on the CRD Map Viewer

Step 8.1: On the layer's detail page in CRD, click **"Map View"** or **"View on Map"**.

You should see: An interactive map showing your 3 flood assessment points near Tanga. The points should be clickable.

Step 8.2: Click on one of the points.

You should see: A popup or sidebar showing the attributes you collected — flood depth, ward name, number of households affected, and damage types. This is your field data, collected in KoboToolbox, now fully integrated into the CRD platform.

Step 8.3: Go back to the CRD homepage (`http://localhost`) and click **"Data"** or **"Datasets"**.

You should see: Your "Flood Damage Assessment" layer listed alongside any other layers you uploaded earlier (like "Mwanza Business Locations"). Anyone who visits your CRD can now find and view this data.

Step 8.4: Try the search: type "flood" in the search box.

You should see: The "Flood Damage Assessment - Tanga 2026" dataset appears in the search results, found by the keywords and title you set.

The complete data pipeline:

1. **Field:** Surveyor collects data with KoboToolbox on a phone (GPS, photos, measurements)
2. **Server:** Data is stored in KoboToolbox with GPS coordinates
3. **Export:** Data manager downloads CSV from KoboToolbox
4. **Transform:** QGIS converts CSV coordinates into geographic points (GeoJSON)

5. **Publish:** GeoJSON is uploaded to CRD with proper metadata
6. **Share:** Anyone can find, view, and download the data through CRD

This entire pipeline uses only free, open-source tools and requires no programming.

Checkpoint

Before proceeding to Practical 7, confirm:

- ☐ **CSV exported:** Downloaded from KoboToolbox with coordinate columns
- ☐ **Points visible in QGIS:** 3 points appear near Tanga on the QGIS map
- ☐ **GeoJSON created:** `flood_assessment_tanga.geojson` saved on your Desktop
- ☐ **Uploaded to CRD:** The layer is visible in your CRD's data catalogue
- ☐ **Metadata set:** Title, abstract, and keywords are filled in
- ☐ **Clickable on map:** Clicking a point in the CRD map viewer shows survey attributes

Outstanding work! You have built a complete data pipeline from field collection to web platform — the core workflow for any SDI. In the final practical, you will create professional map outputs using QGIS and CRD data.

End of Practical 6 — Proceed to Practical 7: Professional Maps with QGIS & CRD Data